Two Independent Single-Loop Voltage Mode Control for a Three-Level Buck Converter

LTspice reconstruction and evaluation

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Abstract

A three-level buck converter with two voltage-mode feedback loops was reconstructed in LTspice to examine output regulation, flying-capacitor balance and startup. At a nominal 66 V input, the prebiased converter regulates the output to 32.00053 V over the final measurement window. Separate initialized tests maintain an approximately 32 V output at 48, 66 and 70 V input and respond to load changes. Zero-energy startup produces excessive current: the incomplete cold-start simulation reaches 580.63 A, while a separate output-reference ramp trial reaches 198.21 A and a flying-capacitor voltage of 67.23 V. During reference ramping, current continues to rise after the output controller reaches its minimum duty because the other loop keeps energy-transfer states active. Integrating the inductor voltage predicts a 25.5383 A increase, agreeing with the measured 25.5384 A. Both 60 V switch banks exceed their blocking ratings. The reconstruction demonstrates prebiased regulation and identifies a startup interaction that requires coordinated energy limitation.

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1 Introduction

Compact display electronics require power supplies whose inductors and capacitors fit within a limited board height. Jeong et al. [1] investigate a three-level buck converter for this problem. Its flying capacitor creates an intermediate switching voltage, reducing the voltage excursions applied to the output inductor under balanced operation. The benefit depends on controlling the capacitor voltage as well as the output.

The published method assigns one voltage-mode feedback loop to each quantity. It avoids the instantaneous peak or valley current measurements used by the current-mode arrangements discussed in the paper and uses the same control configuration on either side of a conversion ratio of one half. The authors evaluate the method with a 500 W-rated hardware prototype.

This study examines how the converter's switching states and reconstructed voltage controllers affect regulation, load response and startup. The analysis connects the simulated voltages and currents to the energy-storage equations, with particular attention to excessive startup current and semiconductor stress.

2 Theoretical background and operating principles

2.1 Buck conversion and energy storage

For an ideal buck converter in periodic continuous conduction, with equal main-switch duty ratios $D_A = D_B = D$, the average conversion relation is

$$V_o \simeq D V_{\mathrm{in}}, \qquad M = \frac{V_o}{V_{\mathrm{in}}}. \tag{1}$$

Here D is the common main-switch duty ratio and M is the conversion ratio. The inductor and capacitor dynamics are

$$L_o \frac{di_L}{dt} = V_{\mathrm{SW}} - V_o, \tag{2}$$

$$C_o \frac{dV_o}{dt} = i_L - i_{\mathrm{load}}, \tag{3}$$

$$C_{\mathrm{Fly}} \frac{dV_{\mathrm{Fly}}}{dt} = i_{C_{\mathrm{Fly}}}. \tag{4}$$

Here V_{SW} is the voltage at the inductor input. Capacitor current is positive into its defined positive terminal. These equations neglect parasitic voltage drops; the switching simulation must account for the modeled nonideal components.

2.2 Power-stage topology

The circuit in figure 1 has four switching positions. The input-side pair connects the input rail to the internal network, while the other devices provide paths to the switching node and return. The flying capacitor spans the internal nodes and transfers charge as the conducting path changes. The output inductor and capacitor filter the switched voltage, and the load receives the resulting DC power.

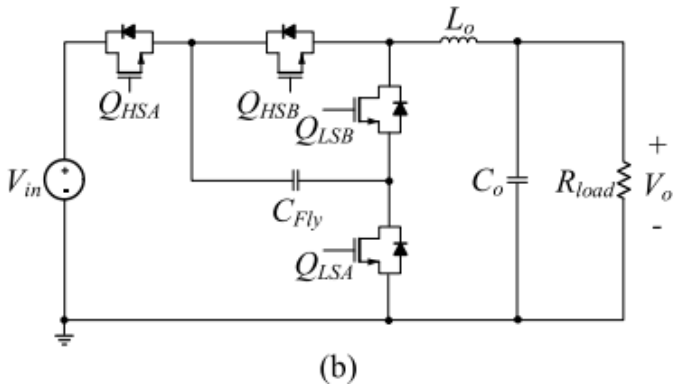


Figure 1: Three-level buck topology from Fig. 1(b) of Jeong et al.[1].

2.3 Switching states and conversion regions

The ideal conducting states in figure 2 give the relationships in table 1, using the switch labels of the paper. Flying-capacitor charge directions assume positive inductor current; reverse current reverses them.

For $M < 0.5$, the sequence is $1 \to 2 \to 3 \to 2$. For $M > 0.5$, Mode 4 replaces Mode 2 in the switching sequence. Substituting each switching voltage into equation (2) gives the current slope in that interval. The ripple pattern therefore differs between the two conversion regions.

Table 1: Ideal operating states, following Section II-A of [1].

Mode	Conducting switches	V_{SW}	Flying-capacitor action
1	$Q_{\mathrm{HSA}}, Q_{\mathrm{LSB}}$	$V_{\mathrm{in}} - V_{\mathrm{Fly}}$	Charges through the inductor path
2	$Q_{\mathrm{LSA}}, Q_{\mathrm{LSB}}$	0	Bypassed in the ideal state
3	$Q_{\mathrm{HSB}}, Q_{\mathrm{LSA}}$	V_{Fly}	Discharges through the inductor path
4	$Q_{\mathrm{HSA}}, Q_{\mathrm{HSB}}$	V_{in}	Bypassed in the ideal state

2.4 Flying-capacitor balance

The intermediate voltages become equal when

$$V_{\mathrm{in}} - V_{\mathrm{Fly}} = V_{\mathrm{Fly}} \quad \Rightarrow \quad V_{\mathrm{Fly}} = \frac{V_{\mathrm{in}}}{2}. \tag{5}$$

This condition makes the mode-1 and mode-3 current slopes equal at a given output voltage. Periodic steady operation also requires volt-second and charge balance:

$$\int_{t}^{t+T_s} (V_{\mathrm{SW}} - V_o)\, dt = 0, \qquad \int_{t}^{t+T_s} i_{C_{\mathrm{Fly}}}\, dt = 0. \tag{6}$$

Under the balanced ideal conditions analyzed in the paper, the switch voltage stresses are approximately $V_{\mathrm{in}}/2$, and the inductor sees two ripple cycles per switching period. If V_{Fly} departs from its target, the two intermediate levels split and the ripple grows. Startup imbalance can raise device stress above this balanced value.

2.5 Two voltage-mode control loops

The published arrangement in figure 3 compares the sensed flying-capacitor voltage with $V_{\mathrm{in}}/2$ and the sensed output with V_{ref}. Compensated errors determine the PWM commands for switching groups A and B, respectively. The carriers have a phase difference of 180°. The paper uses

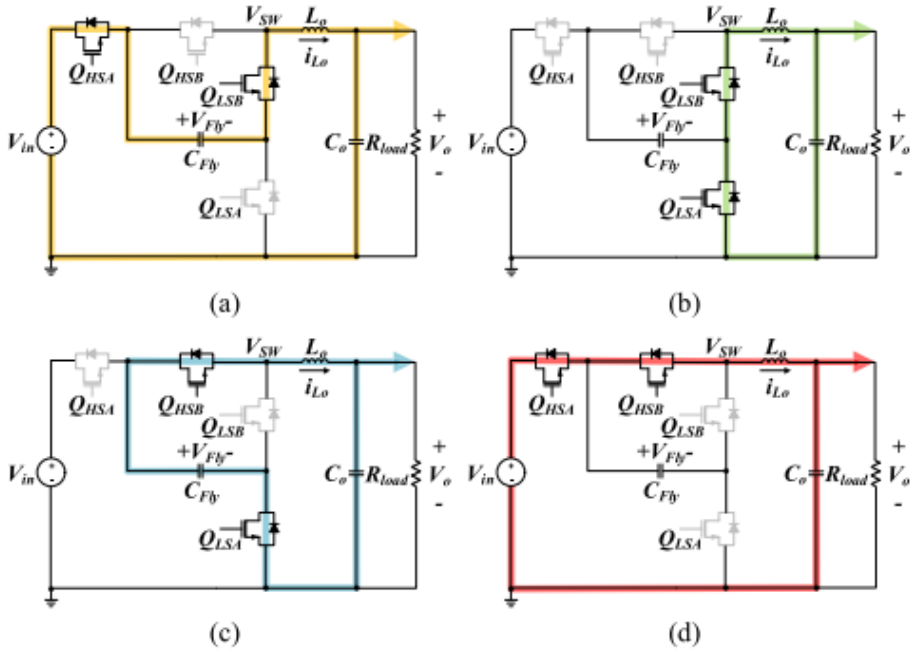


Figure 2: Four ideal conducting modes from Fig. 5 of Jeong et al.[1].

triangular carriers and discusses how carrier shape affects capacitor charge transfer during transients. With sensing gains absorbed into the compensator definitions, a generic reconstruction can be written as

$$e_F(t) = V_{\text{in}}(t)/2 - V_{\text{Fly}}(t), \qquad e_o(t) = V_{\text{ref}}(t) - V_o(t), \tag{7}$$

$$u_A(s) = G_F(s)E_F(s), \qquad u_B(s) = G_o(s)E_o(s), \tag{8}$$

$$D_k(t) = \text{sat}_{[D_{k,\min}, D_{k,\max}]}(u_k(t)), \quad k \in \{A, B\}. \tag{9}$$

Here u_k denotes a normalized duty demand, including any scaling and bias. The two errors have separate compensators, but both commands change the same switching network. Their interaction is therefore important during transients.

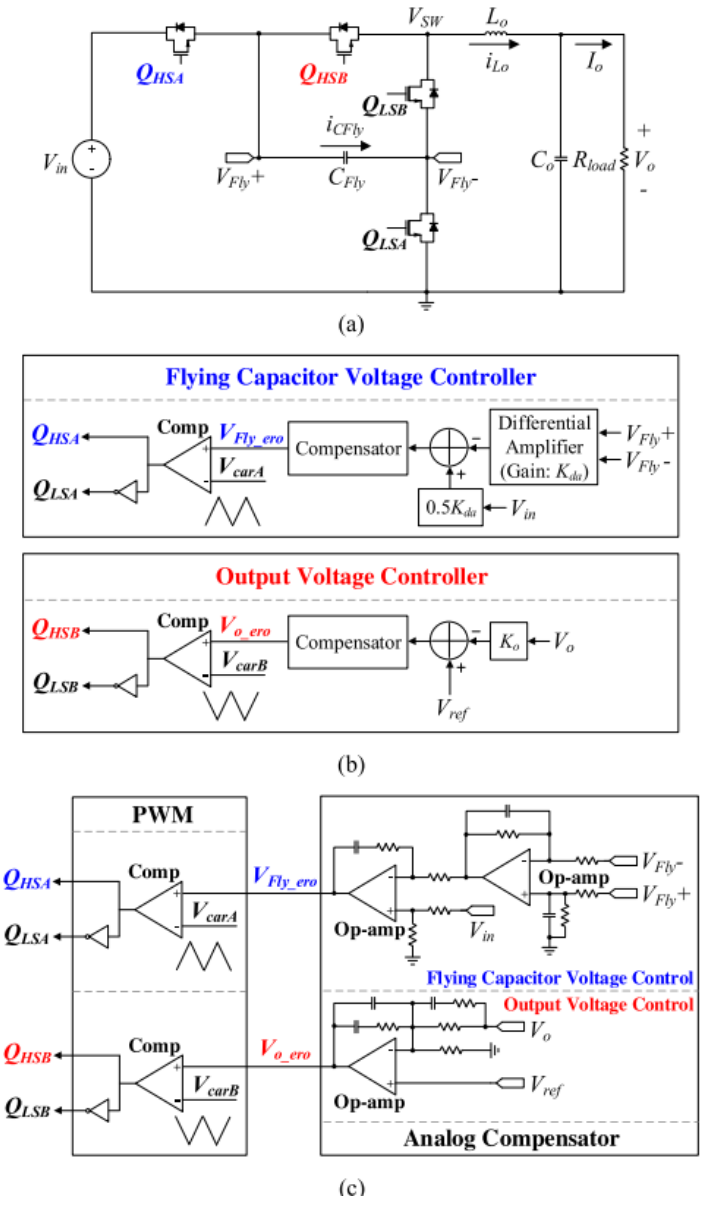


Figure 3: Published voltage-control arrangement from Fig. 9 of Jeong et al.[1].

3 LTspice circuit implementation

3.1 Model parameters and power stage

The converter uses the nominal values in Table 2. The reference-ramp configuration includes a 2.048 ohm load, 3 milliohm flying-capacitor ESR and 13 milliohm output-capacitor ESR.

At the nominal target, the load gives

$$P_o = \frac{V_o^2}{R} = \frac{32^2}{2.048} = 500\,\mathrm{W}, \qquad I_o = \frac{32}{2.048} = 15.625\,\mathrm{A}.$$

Table 2: Nominal converter parameters.

Quantity	Setting
Input / output reference	66 V / 32 V
Flying-capacitor target	33 V at nominal input
Switching frequency	100 kHz per switching group
C_{Fly}, C_o, L_o	20 µF, 500 µF, 1.36 µH
$Q_{\mathrm{HSA}}, Q_{\mathrm{LSA}}$	Two parallel Infineon BSC034N10LS5 devices per position
$Q_{\mathrm{HSB}}, Q_{\mathrm{LSB}}$	Two parallel Infineon BSC027N06LS5 devices per position
Load resistance	2.048 Ω nominal; the load-step tests use 8.192 / 2.73067 ohm
Capacitor ESR	C_{Fly}: 3 mΩ; C_o: 13 mΩ (the ESR comparison also tests zero)
Nominal output power / current	500 W / 15.625 A at 32 V
Case-temperature boundary	Fixed 25 °C

The small inductance gives $di_L/dt = 0.7353\,\mathrm{A/\mu s}$ for each volt across the inductor; even 10 V applied for 1 µs changes current by approximately 7.35 A. Switching-state duration and capacitor imbalance therefore have a large effect on startup current.

The four switching positions contain eight modeled MOSFETs. Parallel devices share the position's applied voltage; they do not halve its voltage stress.

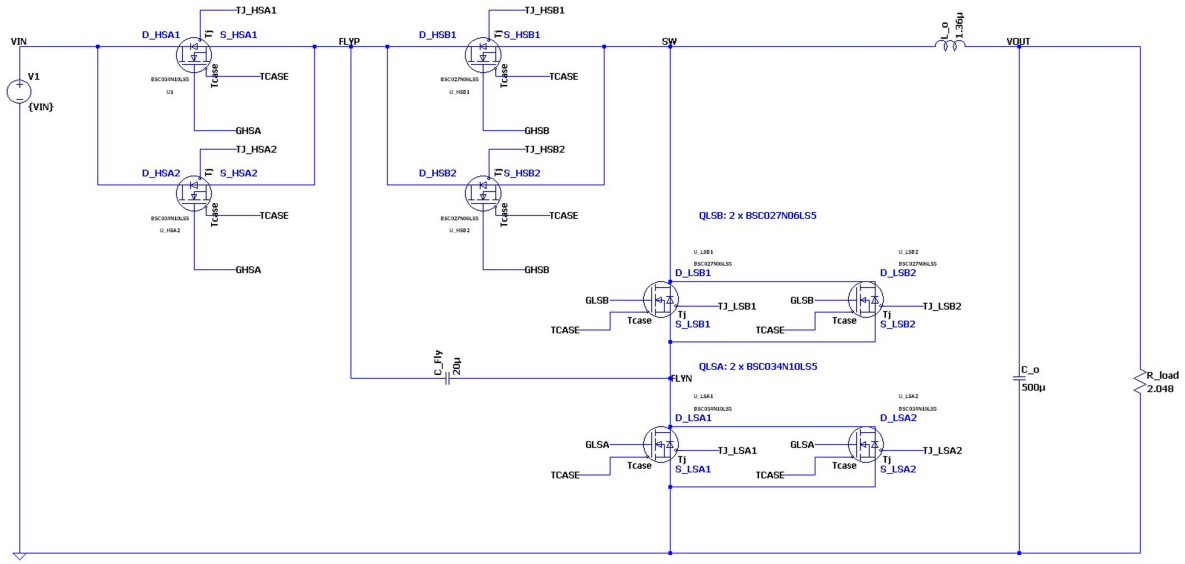


Figure 4: LTspice power stage with parallel MOSFET banks, flying capacitor and output filter.

3.2 Device models and temperature boundary

Manufacturer models represent the electrical behavior of the two Infineon device types. Each custom symbol connects drain, gate, source, junction temperature and case temperature to the corresponding model terminals. The 100 V and 60 V libraries use versions 041121 and 100122, respectively[4, 5]. The case-temperature terminal is fixed at 25 degrees Celsius. Datasheets specify blocking ratings of 100 V for BSC034N10LS5 and 60 V for BSC027N06LS5[2, 3]. Devices in parallel share the bank voltage, so adding a second device increases current capability rather than blocking voltage.

3.3 Gate drive and PWM

Each gate command must be evaluated relative to its own source:

$$V_{GS,k} = V_{G,k} - V_{S,k}. \tag{10}$$

The switching period is $T_s = 1/f_s = 10\,\mu\text{s}$. Complementary commands use a nonoverlap gap to reduce simultaneous turn-on within a switching group. Behavioral sources apply each gate voltage relative to its own source, which is essential for the floating upper switches.

3.4 Reconstructed feedback implementation

The normal initialized feedback configuration uses fixed references

$$e_A = V_{\text{in}}/2 - V_{\text{Fly}}, \qquad e_B = 32 - V_o. \tag{11}$$

The fixed-PWM comparison reuses the nominal feedback record as its controlled trace and uses a separate fixed-pulse-drive run as its open-loop counterpart. The reference-ramp run alone replaces the output error by $e_B = 32\min(t/1\,\text{ms}, 1) - V_o$. This changes the reference, not the compensator coefficients. Both feedback commands use the saved input *parameter* for the bias:

$$D_A = \text{limit}(32/V_{\text{IN,param}} + C_A, 0.02, 0.98), \quad D_B = \text{limit}(32/V_{\text{IN,param}} + C_B, 0.02, 0.98). \tag{12}$$

During source application in the reference-ramp run, the fixed input parameter differs from the instantaneous input-node voltage.

The compensators use RC states to implement their transfer functions. A unit state capacitor driven by ke, with a $100\,\Omega$ leakage resistor, gives $x/e = k/(s + 0.01)$. A buffered low-pass state has $f = x/(1 + \tau s)$; the implemented lead combination $ax + (1 - a)f$ therefore equals $(1 + a\tau s)x/(1 + \tau s)$. Applying this to the two subcircuits gives

$$G_A(s) = \frac{10.4593927209}{s + 0.01}\,\frac{1 + 7.17383307414\tau_A s}{1 + \tau_A s}, \tag{13}$$

$$G_B(s) = \frac{1661.90546898}{s + 0.01}\,\frac{1 + 6.18140633065\tau_B s}{1 + \tau_B s}\,\frac{1 + 12.3628126613\tau_C s}{1 + \tau_C s}, \tag{14}$$

where $\tau_A = 53.051\,65\,\mu\text{s}$, $\tau_B = 3.978\,874\,\mu\text{s}$ and $\tau_C = 1.989\,437\,\mu\text{s}$. The corresponding filter poles are approximately 3, 40 and 80 kHz; the lead zeros are approximately 418.2 Hz, 6.471 kHz and 6.471 kHz. These coefficients define the reconstructed controllers. All internal capacitors specify zero initial state. No explicit clamp-dependent integrator reset or back-calculation appears in this library, so output saturation does not itself implement anti-windup.

The two 100 kHz triangular carriers are complementary. Source-referenced gate expressions use a `tanh` transition with scale 0.001 and a 0.01 command gap. The gap sets command nonoverlap in the behavioral PWM implementation.

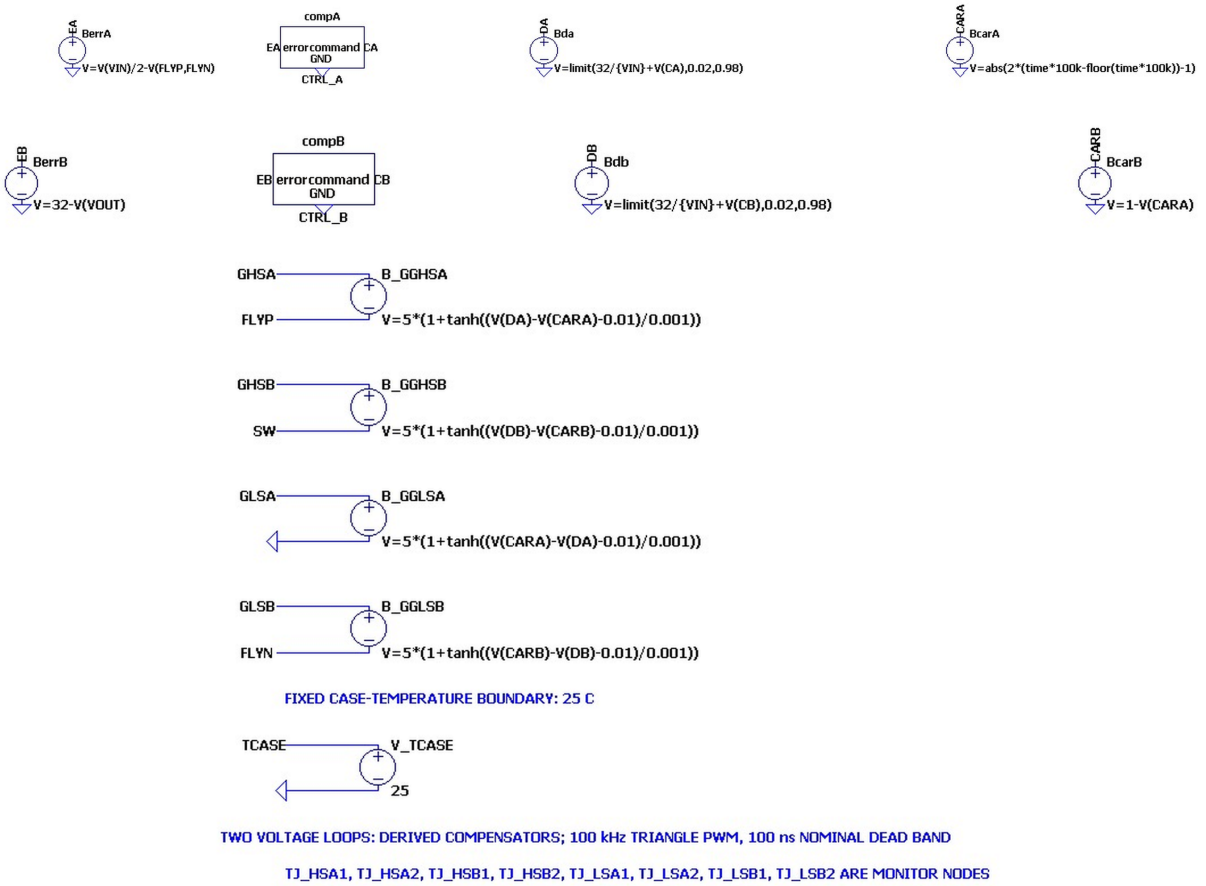


Figure 5: LTspice flying-capacitor loop (top), output-voltage loop (middle) and source-referenced gate-drive expressions (bottom).

4 Simulation Setup and Methodology

LTspice simulations use the manufacturer MOSFET models and the behavioral control implementation described above. Nominal input voltage is 66 V, the output reference is 32 V and the switching frequency is 100 kHz. Separate tests vary input voltage, output-capacitor ESR, load and initial stored energy. The analysis uses the original transient results without rerunning or changing the circuits.

4.1 Operating conditions and initial stored energy

Fixed-PWM simulations examine switching and open-loop startup over 0–450 microseconds. Initialized feedback tests examine recovery through 1.5 ms, constant-input operation through 3 ms, and load changes at 3 and 4 ms. Input-voltage cases are 48, 66 and 70 V. The nominal load is 2.048 ohm; load-change tests switch between 8.192 and 2.73067 ohm.

Zero-energy startup specifies zero initial storage energy and the LTspice `startup` option. Its requested duration is 400 microseconds, with results available through 58.273 microseconds. A separate trial ramps the output reference over 1 ms and is observed through 100 microseconds. The detailed startup-current analysis uses that same reference-ramp trial.

A prebiased simulation begins with energy already stored in one or more reactive elements. The ideal stored energy is

$$E_0 = \tfrac{1}{2}L_o i_L^2(0) + \tfrac{1}{2}C_o V_o^2(0) + \tfrac{1}{2}C_{\mathrm{Fly}} V_{\mathrm{Fly}}^2(0). \tag{15}$$

The nominal feedback recovery test specifies $V_o(0) = V_{\mathrm{Fly}}(0) = 30$ V and $i_L(0) = 14.6484$ A, with zero initial controller states. It uses `uic`, which skips the operating-point solve. The `startup` option instead applies independent sources over the first 20 microseconds[6]. Initial energy and source application therefore differ between the recovery and cold-start tests.

4.2 Measurements and observation windows

For an interval $[t_a, t_b]$, define

$$\overline{x} = \frac{1}{t_b - t_a} \int_{t_a}^{t_b} x(t)\, dt, \tag{16}$$

$$\varepsilon_o = \overline{V_o} - V_{\mathrm{ref}}, \qquad \varepsilon_F = \overline{V_{\mathrm{Fly}}} - \overline{V_{\mathrm{in}}}/2, \tag{17}$$

$$x_{\mathrm{pp}} = \max_{[t_a,t_b]} x - \min_{[t_a,t_b]} x. \tag{18}$$

Means are time-weighted trapezoidal averages, with linear interpolation at measurement-window endpoints. Ripple is the maximum minus minimum over the stated interval; startup peaks are reported separately. The nominal recovery window is 1.45–1.50 ms, and the constant-input window is 2.95–3.00 ms. Load-response analysis covers 2.8–4.8 ms. Differential node voltages determine inductor voltage, gate-to-source voltage and drain-to-source stress. A 5 V gate threshold is used only to classify commanded switching states.

5 Simulation results and discussion

5.1 Switching States and PWM Operation

The source-referenced gates in figure 6 show the commanded switching combinations. Near $M = 0.5$, balanced flying-capacitor voltage brings the two intermediate switching levels close together. The selected interval consequently does not show three clearly separate 0/33/66 V plateaus.

5.2 Open-Loop Performance and Semiconductor Stress

The baseline startup and ESR comparison in table 3 show large current excursions. In equation (2), the small L_o makes the current slope sensitive to the instantaneous voltage difference. Adding ESR provides damping in the reported comparison, but the remaining excursions prevent a satisfactory startup assessment. Both ESR-comparison logs contain tolerance-relaxation warnings, which limit confidence in numerical convergence even when a waveform is available.

Table 3: Output-capacitor ESR comparison over 0–450 microseconds.

Output-capacitor ESR	$i_{L,\max}$ (A)	$i_{L,\min}$ (A)	$V_{o,\max}$ (V)
13 mΩ	391.734	-133.529	43.662
Zero	432.963	-218.867	46.679

The baseline open-loop flying-capacitor peak is 67.547 V. The later open-loop interval, 380 µs to 440 µs, still has output voltage between 31.575 V and 32.426 V; the response is therefore not fully settled.

Peak drain-source voltages are 68.169, 67.859, 66.551 and 68.982 V for HSA, HSB, LSB and LSA, respectively. The 60 V HSB and LSB banks exceed their ratings by 7.859 V (13.1%) and

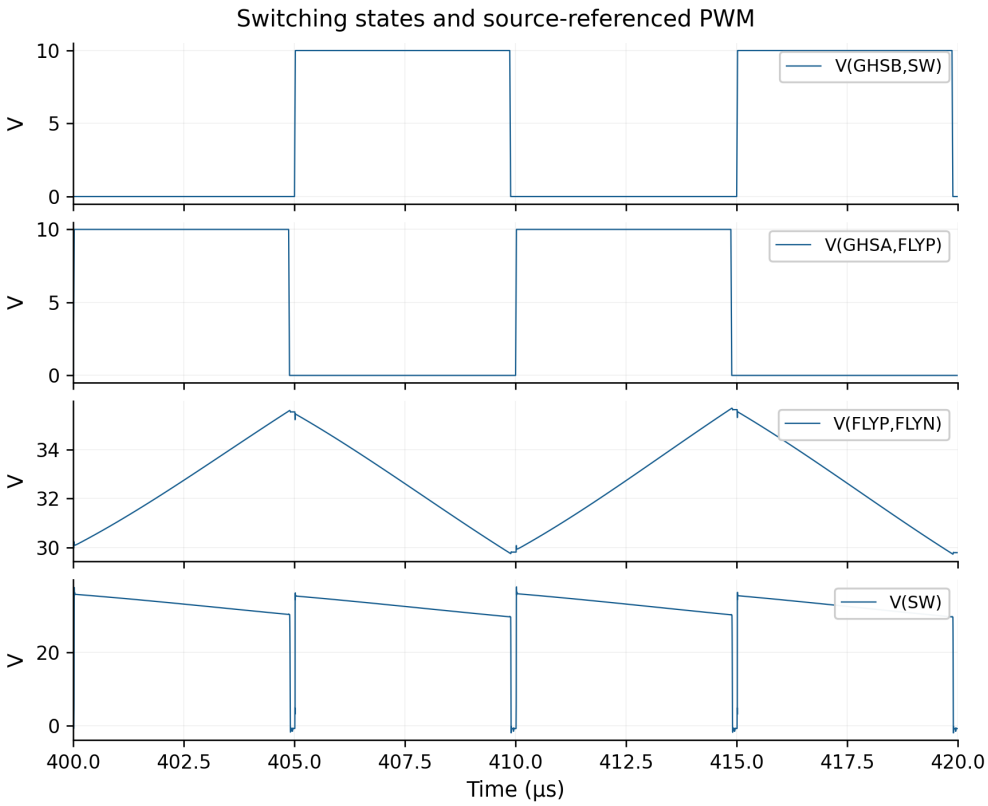


Figure 6: Source-referenced gate commands and switching-node voltage under fixed PWM at nominal input.

6.551 V (10.9%). The 100 V banks remain below their positive blocking ratings. The imbalance therefore removes the expected half-input stress advantage during startup.

Relative to 13 milliohm ESR, zero output ESR increases the positive current peak by 41.229 A (10.5%), reverse-current magnitude by 85.338 A (63.9%) and output peak by 3.017 V. ESR affects both damping and terminal voltage through $R_{\mathrm{ESR}} i_C$. Even with ESR, the current excursions remain excessive. Figures 7–9 show the associated stress and startup responses.

5.3 Closed-Loop Voltage Regulation

The nominal feedback run begins with specified storage-element initial values $V_{\mathrm{Fly}} = 30\,\mathrm{V}$, $V_o = 30\,\mathrm{V}$ and $i_L = 14.6484\,\mathrm{A}$, using `uic`. Thus the output starts below its 32 V target and the flying capacitor below its 33 V target, while both controller internal states start at zero. Both positive errors initially change the duty commands around the feedforward bias. Because the loops act through the same switching network, their first response also redistributes stored capacitor energy. The saved record contains a current peak of 75.876 A, a flying-terminal-voltage peak of 35.520 V and an output peak of 32.926 V; the initial response includes a substantial current excursion.

Later, the output approaches the reference while current supplies the resistive load. Over the exact final 1.45 ms to 1.50 ms window, the time-weighted output mean is 32.000 53 V with 54.823 mV peak-to-peak ripple. Mean inductor current is 15.624 48 A, consistent with the nominal 15.625 A load; its ripple is 4.4012 A. Flying-terminal voltage averages 32.844 64 V, or −0.155 36 V relative to 33 V, with 3.9122 V peak-to-peak variation. The output is regulated closely, but the capacitor average is still approaching its target and the ripple remains substantial. These measurements describe the final 50 microseconds of the recovery test.

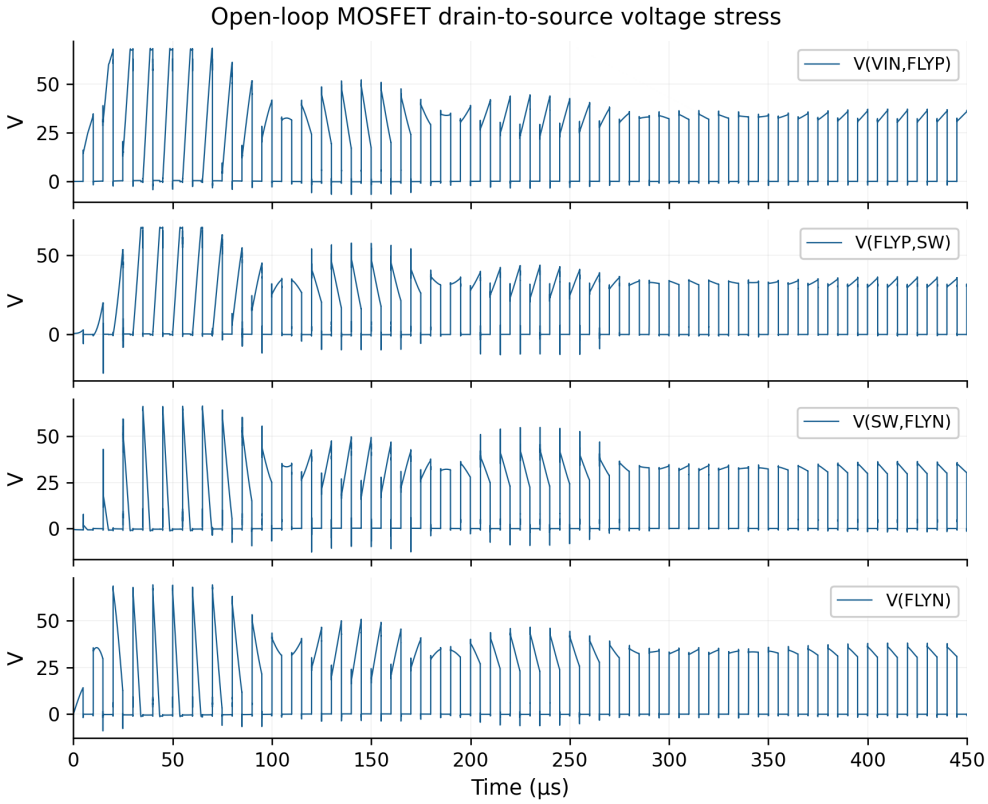


Figure 7: MOSFET drain-to-source voltages during open-loop startup.

The output last leaves a 32 ± 0.32 V (1%) band at approximately 94.39 microseconds and remains inside through 1.5 ms. This is the recovery time observed within the simulated interval.

5.4 Regulation Under Different Input Voltages

The operating-point measurements in table 4 indicate close average output regulation. The ratios $32/48 > 0.5$ and $32/66, 32/70 < 0.5$ place the tests in both conversion regions defined in section 2.3. The capacitor means track their respective balance targets; ripple is evaluated separately below.

Table 4: Prebiased operating points over 2.95–3.00 ms.

V_{in} (V)	$\overline{V_o}$ (V)	$\overline{V_{\text{Fly}}}$ (V)	$V_{o,\text{pp}}$ (V)
48	32.00000	23.999	0.263
66	32.00007	32.979	0.052
70	32.00005	34.985	0.139

The expected feedback outcome is a nearly unchanged output mean with flying voltage following half the input. In the exact same 2.95–3.00 ms window, the current means are 15.62510, 15.62505 and 15.625 04 A for 48, 66 and 70 V, respectively. Their current ripples are 20.4704, 4.1245 and 10.6639 A; flying-terminal ripples are 2.6448, 3.9042 and 3.7040 V. The similar output means are accompanied by different switching-current demands. In particular, the 48 V case has the largest current ripple even though its flying-voltage ripple is the smallest.

The ideal conversion ratios are 0.6667, 0.4848 and 0.4571. The 48 V case operates above one half, while 66 and 70 V operate below it. Different state durations change the voltage applied to the

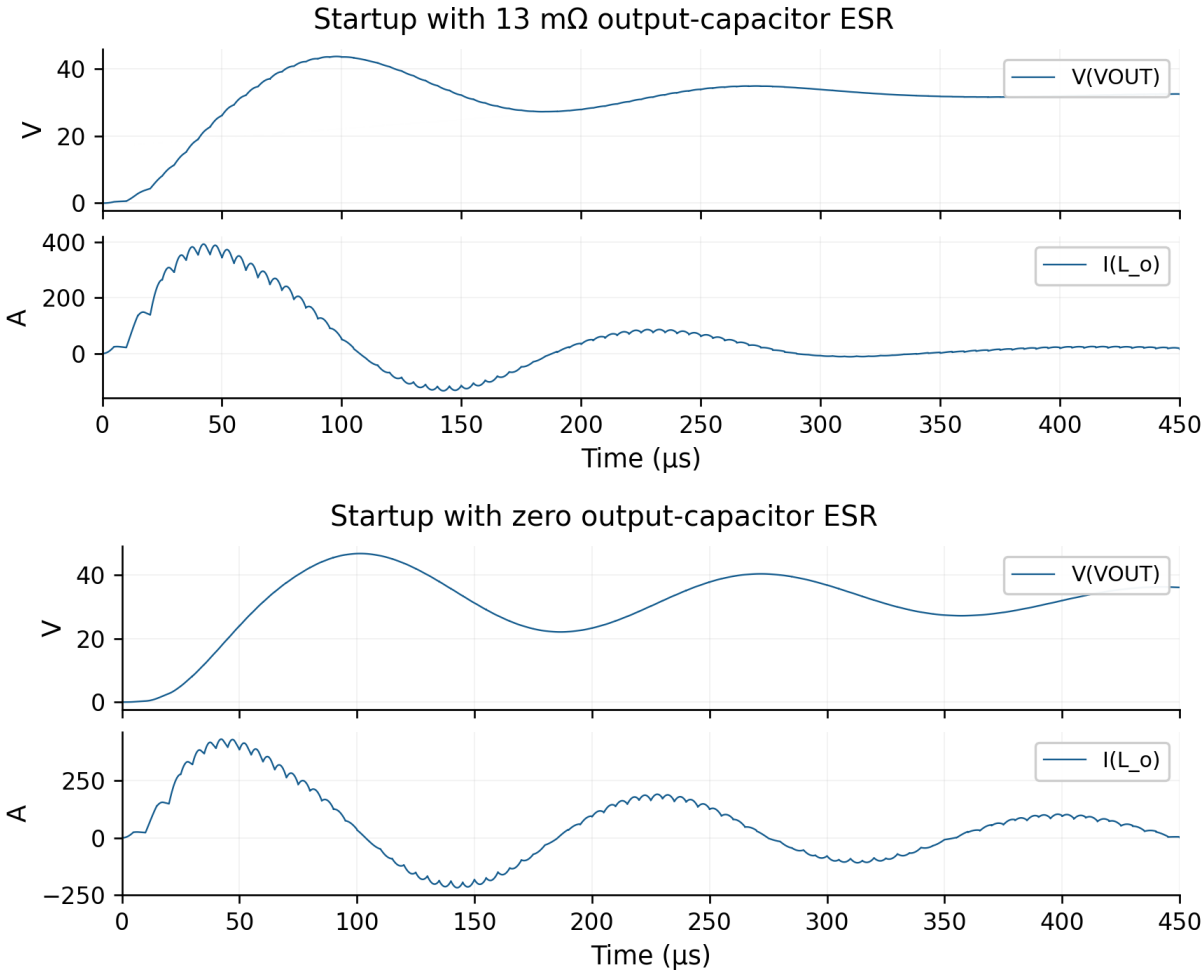


Figure 8: Open-loop startup with 13 milliohm output-capacitor ESR (top) and zero ESR (bottom).

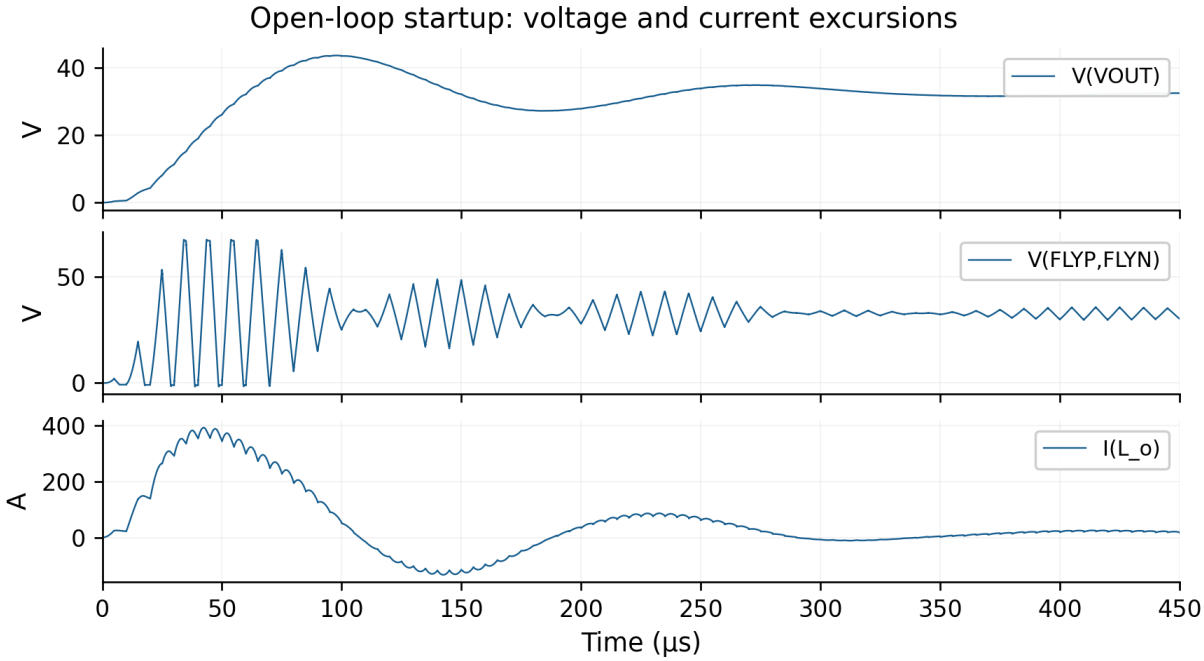


Figure 9: Output voltage, flying-capacitor voltage and inductor current during open-loop startup.

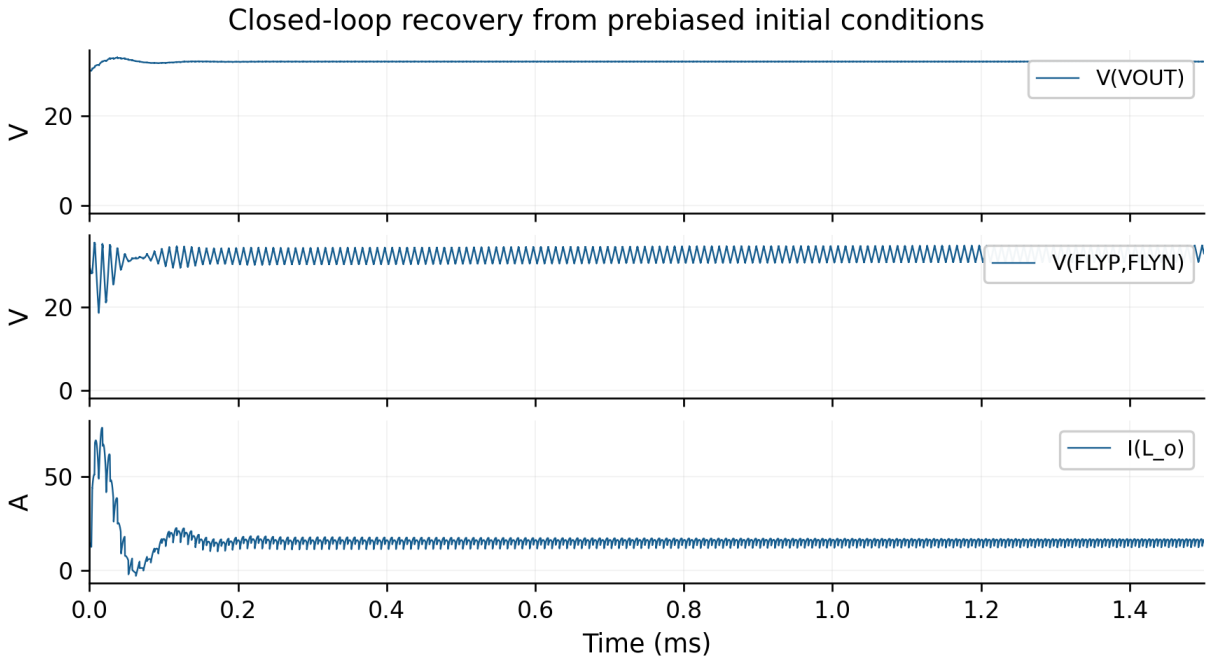


Figure 10: Closed-loop recovery at 66 V input from prebiased storage elements.

inductor. At 66 V, the balanced intermediate voltage is close to the 32 V output, which helps explain the smaller current ripple. These are separate constant-input tests of line regulation.

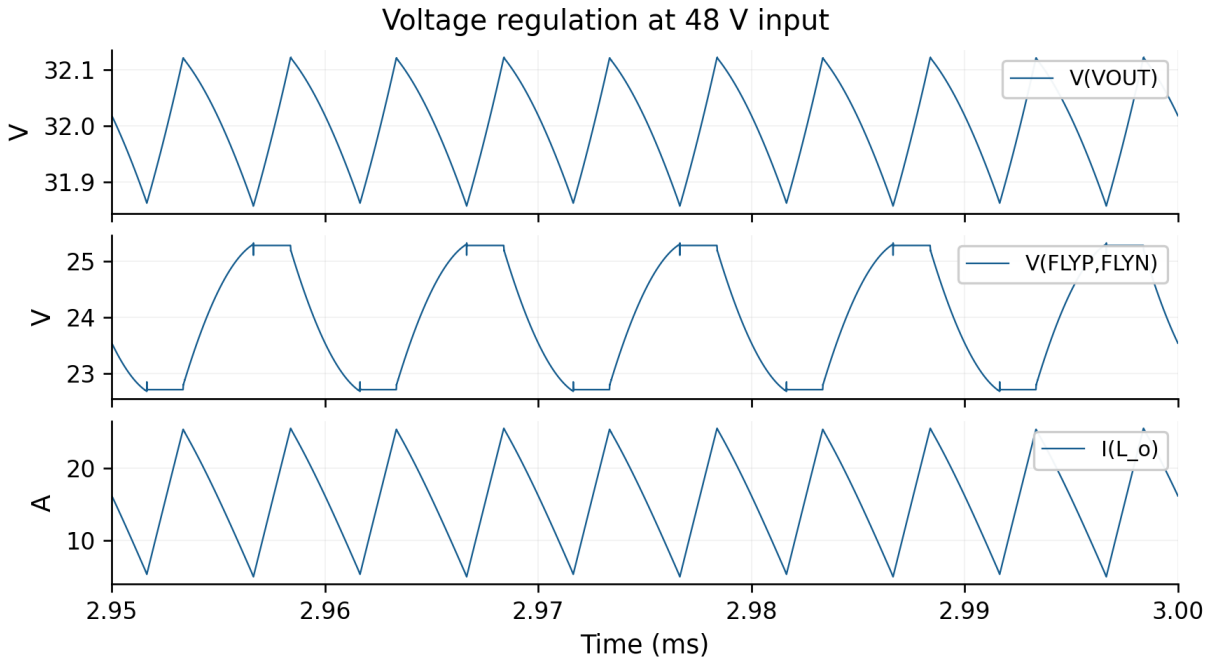


Figure 11: Output and flying-capacitor voltages with inductor current at 48 V input.

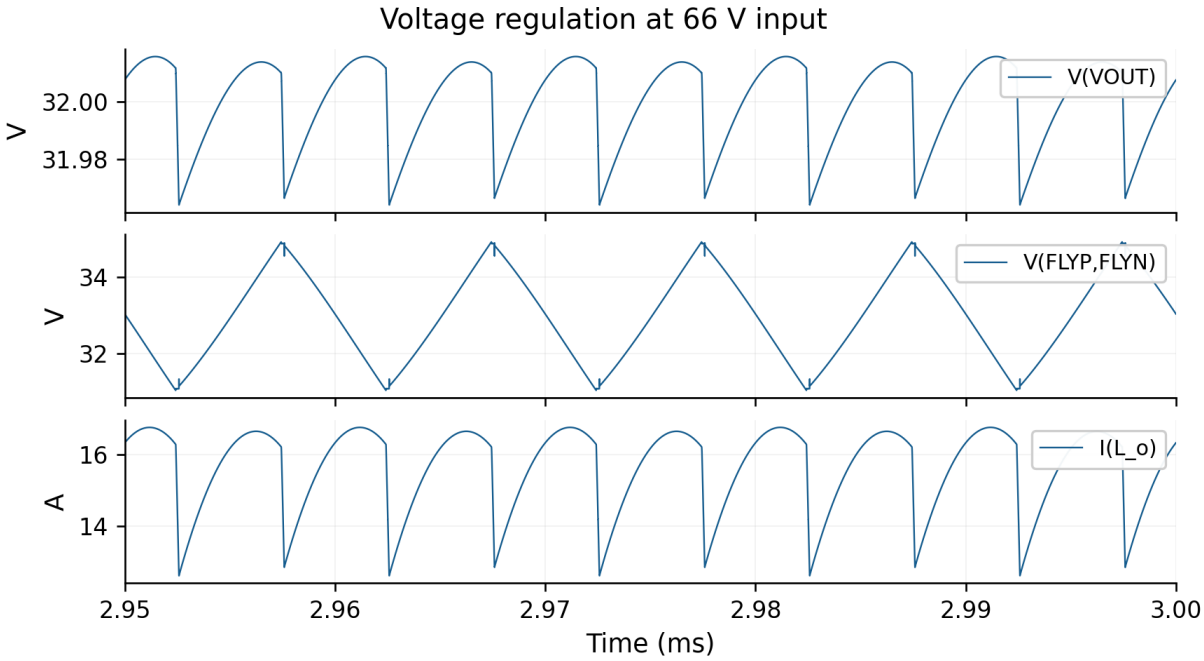


Figure 12: Output and flying-capacitor voltages with inductor current at 66 V input.

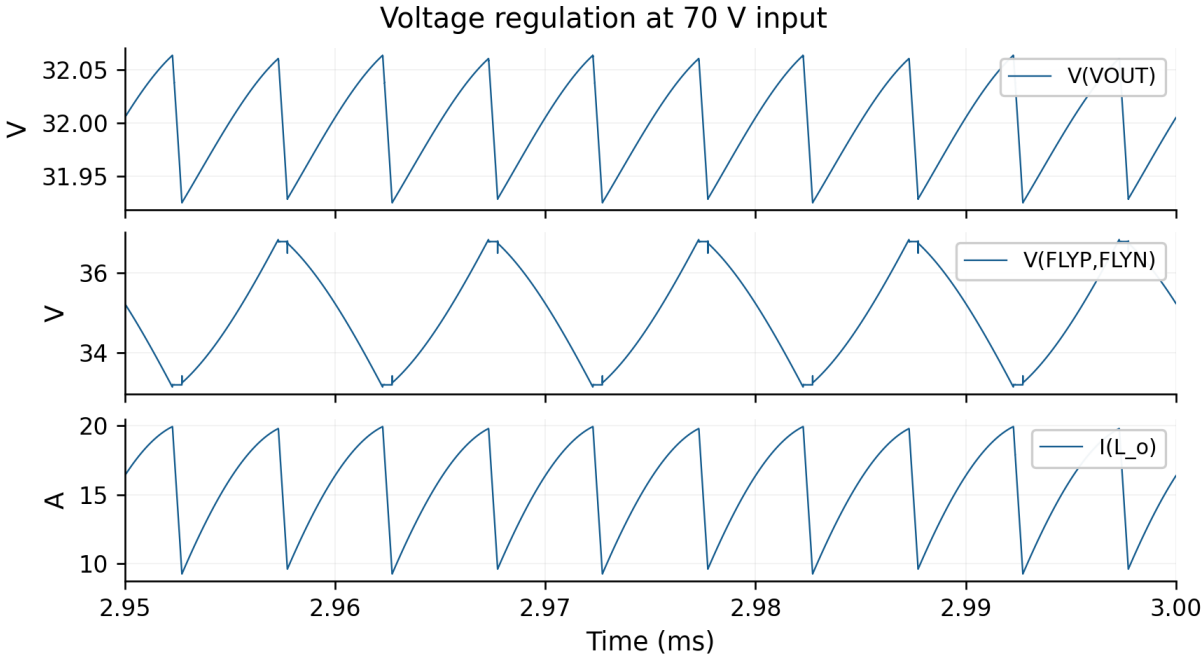


Figure 13: Output and flying-capacitor voltages with inductor current at 70 V input.

5.5 Dynamic Response to Load Changes

A load increase initially draws additional current from the output capacitor until the inductor current responds, as described by equation (3). The reported tests change the load from 25% to 75% and back. Their output deviations are approximately 0.17 V to 0.32 V around the reference, with switching ripple included in the extrema. The saved load resistor changes from 8.192 to 2.7306667 ohm at 3 ms and returns at 4 ms. The corresponding nominal currents at 32 V are 3.90625 and 11.71875 A; actual current follows the instantaneous output voltage. At the 3 ms load increase, the immediate current deficit is approximately 7.8125 A. Ignoring the ESR step initially gives $dV_o/dt \approx -7.8125/(500\,\mu\text{F}) = -0.015\,625\,\text{V}/\text{ts}$. The 13 mΩ ESR also produces a direct voltage change of order 0.102 V. The feedback then raises the current demand. On load release, excess inductor current instead charges the output capacitor, causing the opposite voltage excursion. These estimates explain the direction of the response; the saved switching waveform determines the actual transient.

For 48, 66 and 70 V input, the minimum output over 3–4 ms is 31.67668, 31.82610 and 31.784 58 V, respectively. The maximum after load release over 4–4.8 ms is 32.31843, 32.18495 and 32.231 87 V. Thus the largest observed deviations in these two windows are approximately 1.01%, 0.58% and 0.72% of 32 V. Switching ripple is included. Mean inductor current over 3–4 ms is 11.71444, 11.71615 and 11.717 73 A; over 4–4.8 ms it is 3.90867, 3.90840 and 3.907 13 A, consistent with the changed load levels. The flying-capacitor voltage also changes as the balancing loop responds to the altered switching demand.

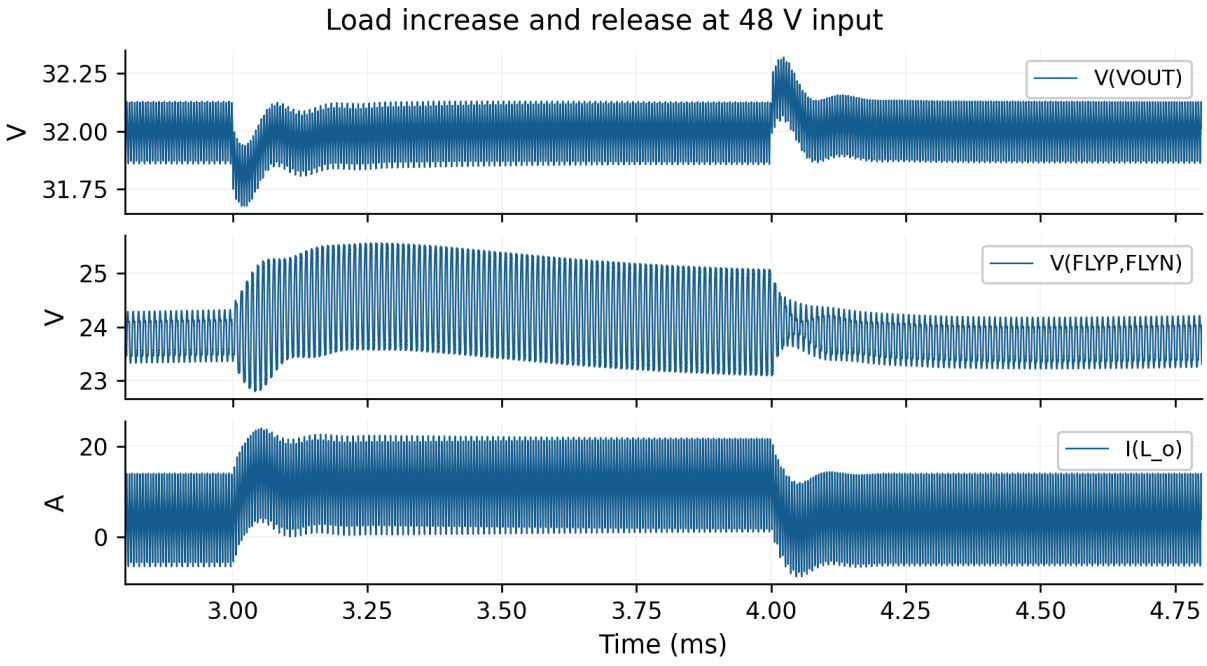


Figure 14: Voltage and current response to load changes at 48 V input.

5.6 Fixed PWM versus Closed-Loop Control

The comparison evaluates fixed pulse drives against the existing nominal feedback record at 66 V and 2.048 Ω. Both schematics specify the same storage-element initial conditions (30 V on both capacitors and 14.6484375 A in the inductor), `uic`, 1.5 ms stop time and 30 ns maximum step. The command implementations differ: open-loop PULSE sources versus closed-loop triangular-

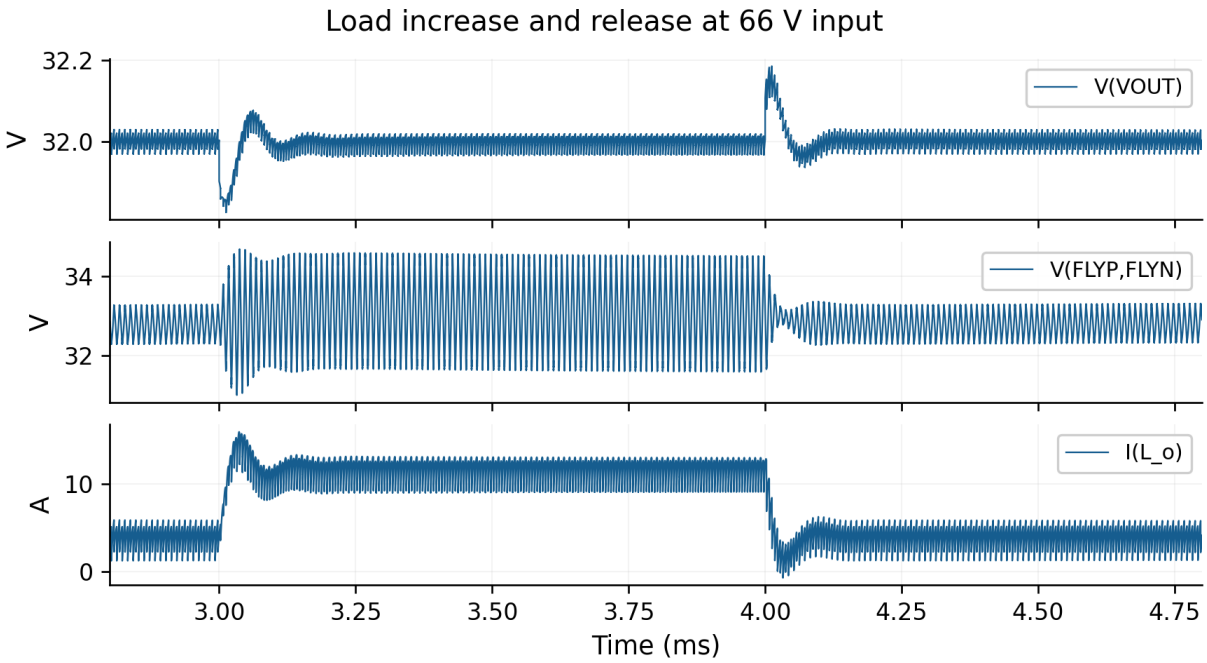


Figure 15: Voltage and current response to load changes at 66 V input.

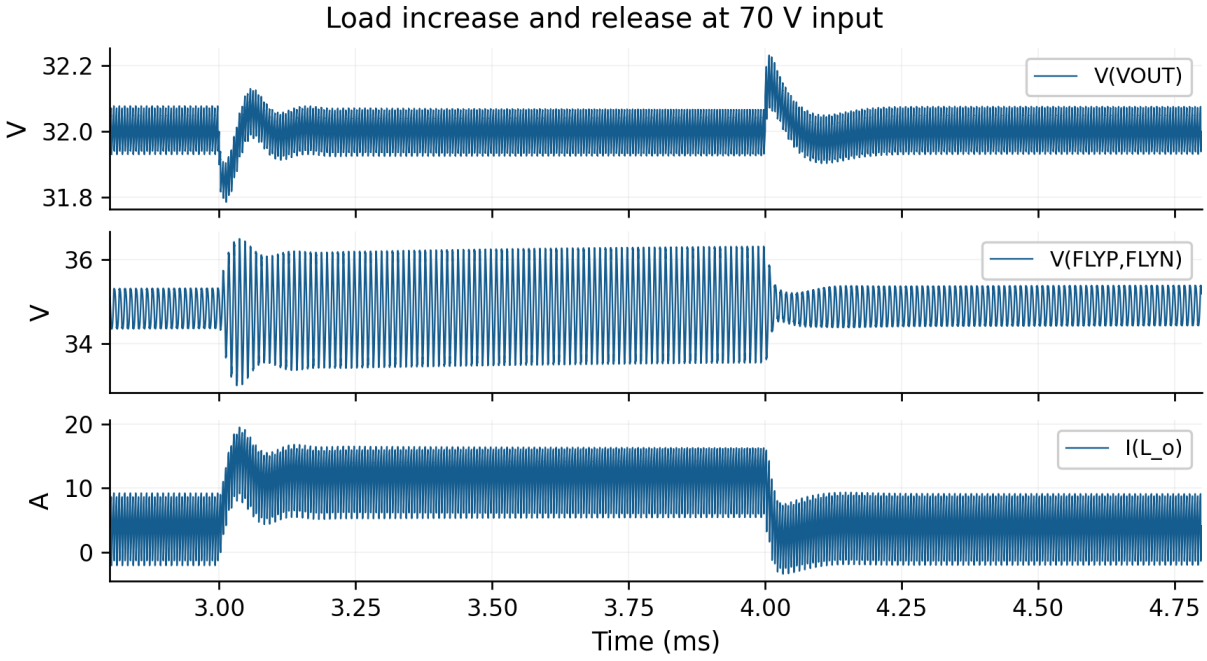


Figure 16: Voltage and current response to load changes at 70 V input.

carrier behavioral sources. The comparison therefore evaluates the two saved implementations, with both feedback and pulse-generation differences.

Table 5: Matched fixed-PWM and feedback measurements. Late window: 1.45–1.50 ms. Peaks: complete saved 0–1.50 ms record.

Metric	Fixed PWM	Feedback
Late mean output (V)	32.18633	32.00053
Mean error from 32 V (V)	+0.18633	+0.00053
Late output ripple (V pp)	0.08282	0.05482
Peak output (V)	33.39848	32.92607
Overshoot above 32 V (%)	4.370	2.894
Late flying-terminal mean (V)	31.99919	32.84464
Late current mean (A)	15.71619	15.62448
Late current ripple (A pp)	6.4483	4.4012
Early/full-record current peak (A)	47.9637	75.8761

Feedback reduces late output error by approximately 0.185 80 V, output ripple by 33.8% and observed output overshoot by 1.476 percentage points. The fixed-duty circuit cannot correct its modeled loss and switching-timing offset, whereas feedback adjusts the command to compensate for voltage error. The higher early feedback current peak is a tradeoff: the smaller output error and overshoot come with a larger initial current excursion.

Using the same 32 ± 0.32 V band, the final excursions occur at 0.28499 ms with fixed PWM and 0.09439 ms with feedback. Both remain inside through 1.50 ms. Feedback therefore gives faster output recovery within this band, alongside the larger early current peak.

5.7 Cold-Start and Soft-Start Behavior

The zero-energy cold-start test reaches 580.63 A at 47.729 microseconds. The simulation ends at 58.273 microseconds of the requested 400 microseconds. During the current rise, the output-loop duty is near its upper clamp of 0.98, consistent with a large positive output error. Capacitor imbalance and the applied input ramp also affect the transient.

A separate schematic copy applies an output-reference ramp:

$$V_{\mathrm{ref},o}(t) = 32\,\mathrm{V}\,\min\left(\frac{t}{1\,\mathrm{ms}}, 1\right). \tag{19}$$

Table 6 summarizes the reference-ramp response. Its observed peak current is lower than in the incomplete cold-start test, but the two durations and startup conditions differ. The reference-ramp trial still has excessive current, reverse current and flying-capacitor overvoltage.

Table 6: Reference-ramp observations over 0 ţs to 100 ţs.

Quantity	Observation
Inductor current	Maximum 198.21 A at about 25.91 ţs; minimum −76.43 A at about 72.90 ţs
Output voltage	Peak 10.69 V; final 7.71 V
Flying-capacitor voltage	Peak 67.23 V; final 40.03 V
Duty command D_B	Reaches the 0.020 lower clamp by about 18.10 ţs
Solver behavior	Gmin stepping required after direct Newton iteration failed

At the end of this interval, equation (19) gives a reference of only 3.2 V. Failure to reach the final nominal output during the short run is therefore expected; the adverse evidence is excessive current, capacitor overvoltage, and poor reference tracking. An output-reference ramp alone did not meet the startup objective in this configuration.

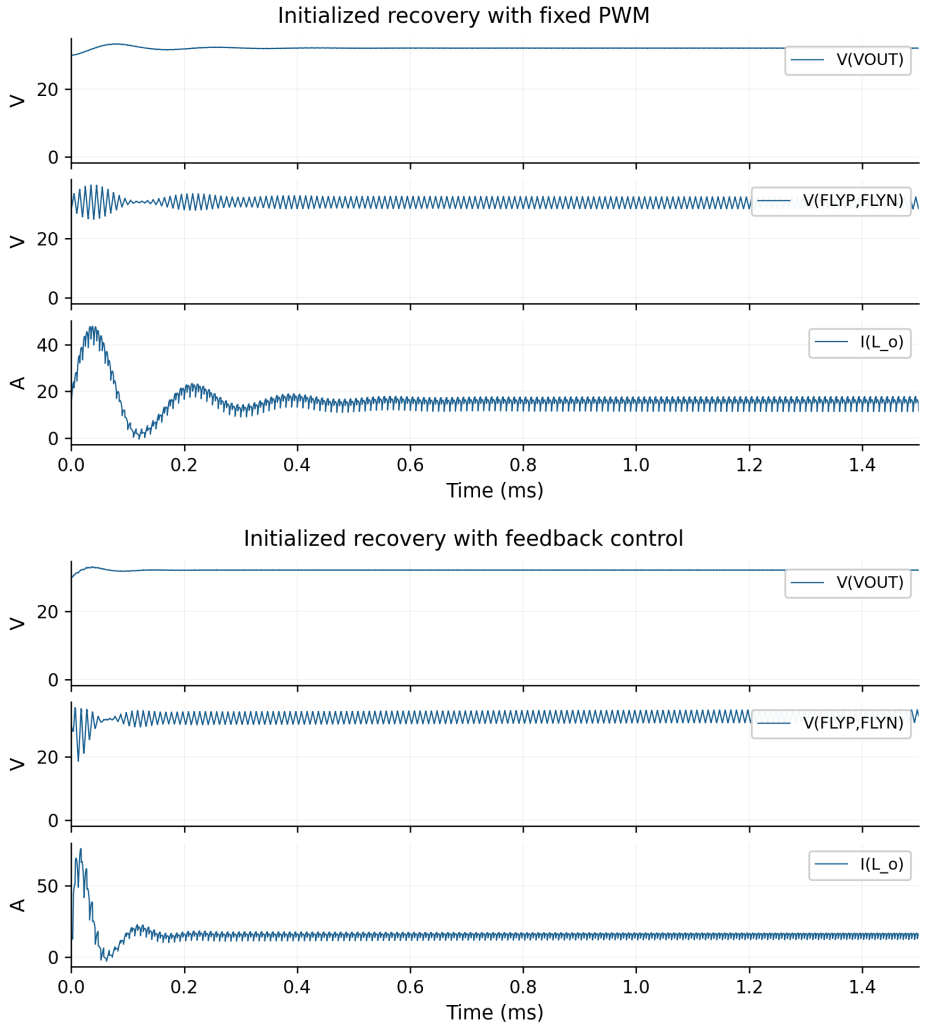


Figure 17: Prebiased recovery with fixed PWM (top) and closed-loop control (bottom) at 66 V input.

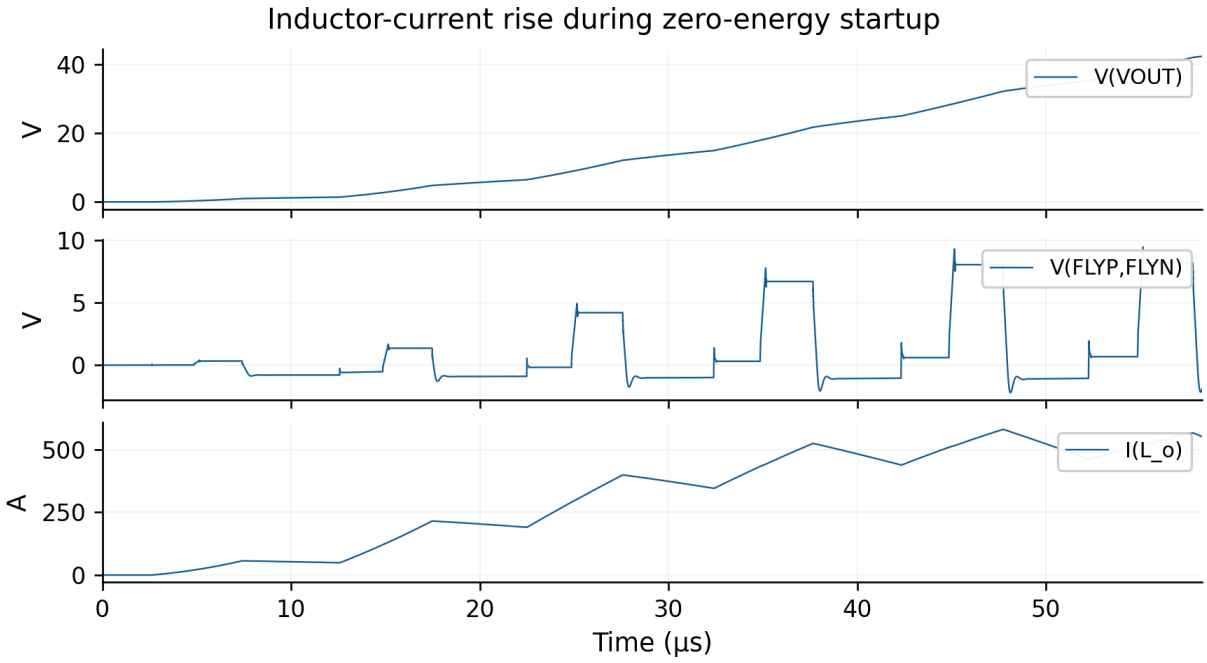


Figure 18: Inductor current and capacitor voltages during incomplete zero-energy cold startup.

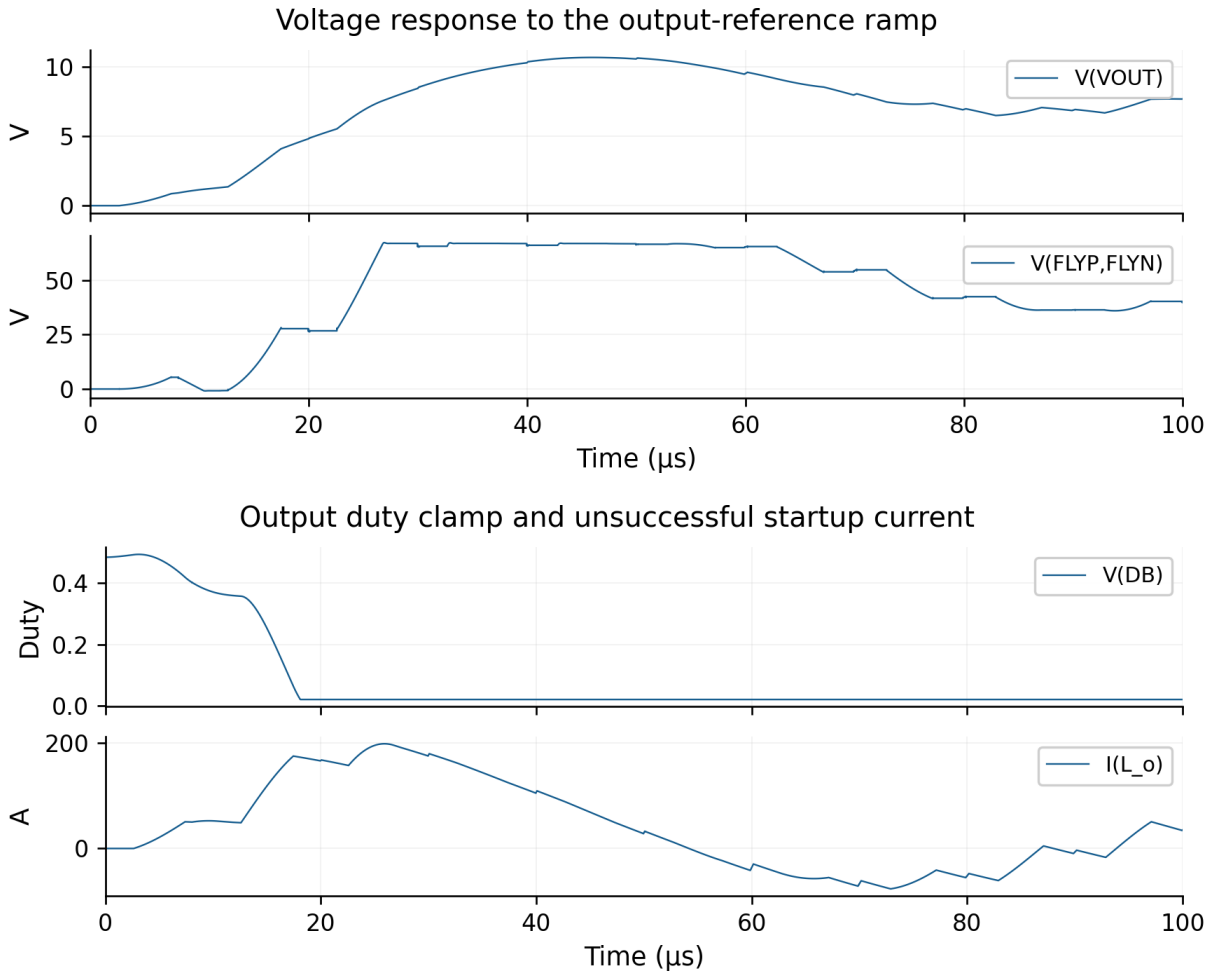


Figure 19: Output and flying-capacitor voltages (top), duty command and inductor current (bottom) with a 1 ms reference ramp.

5.8 Analysis of Excessive Startup Current

The reference-ramp startup is examined in detail using 66 V nominal input, zero specified storage-element initial conditions and a 1 ms reference ramp to 32 V. The independent supply is also subject to the saved `startup` input application.

Applied inductor voltage. At 18.1013 µs, D_B first reaches 0.02 and i_L is 172.674 A. Current subsequently peaks at 198.212 A at 25.9107 µs. Over this interval the measured increase is 25.5384 A. Integrating the saved differential voltage gives

$$\frac{1}{1.36\,\mu\text{H}} \int_{18.1013\,\mu s}^{25.9107\,\mu s} (V_{\text{SW}} - V_o)\,dt = 25.5383\,\text{A}. \tag{20}$$

The mean applied voltage is 4.4475 V. The close agreement confirms that positive net inductor volt-seconds account for the recorded current rise, even though the instantaneous voltage also has negative intervals.

Commanded switching states. With source-referenced gate voltages above 5 V treated as ON, Mode 1 occupies approximately 42.9% of the interval between the D_B clamp and the current peak. Mode 2 occupies 52.0%, Mode 3 about 1.3%, and unclassified/dead-time commands about

3.8%; Mode 4 is absent. These duration estimates use the left stored sample for each time interval. QLSB is commanded ON for about 96.2% and QHSA for 42.9%. Mode 1 therefore continues despite minimum D_B. Its sampled average inductor voltage is approximately 16.78 V. The classification describes gate commands; diode conduction can differ during transitions.

Flying-capacitor charging. Flying-capacitor current is estimated from terminal voltage using the capacitor model, with $C = 20\,\mu\text{F}$ and $R_{\text{ESR}} = 3\,\text{m}\Omega$:

$$V_{\text{Fly,terminal}} = u_C + R_{\text{ESR}} i_C, \qquad i_C = C\frac{du_C}{dt}. \tag{21}$$

Integrating this RC relation with zero initial ideal-capacitor voltage and piecewise-linear terminal voltage gives approximately 1.151 millicoulomb of accumulated charge from 15 microseconds to the 67.2256 V terminal-voltage peak at 26.9002 microseconds. Current is positive from FLYP to FLYN. This is a model-derived estimate; capacitor branch current was not directly recorded.

Controller interaction. At the D_B floor, $D_A = 0.49703$; at the current peak it is still 0.49491. The flying-capacitor loop therefore continues to request substantial switching after the output loop reaches its minimum. The measured commands, gate combinations and volt-second integral support a mechanism in which a minimum output-loop duty leaves energy-transfer states active.

Switch stress. The complete saved 0 µs to 100 µs record gives the following differential-voltage extrema. Ratings are the stated nominal bank ratings; each parallel device sees its bank voltage.

Table 7: MOSFET-bank voltage stress during reference ramping.

Bank	Rating (V)	Minimum V_{DS} (V)	Maximum V_{DS} (V)
QHSA	100	-5.275	43.064
QHSB	60	-20.251	67.704
QLSB	60	-12.491	67.109
QLSA	100	-9.671	43.582

Both 60 V banks exceed their nominal blocking ratings.

The current rise follows directly from the positive average inductor voltage: $di_L/dt = V_L/L$. The minimum output duty leaves Mode 1 active because the flying-capacitor command remains near 0.495. Coordinating the two commands during startup is therefore a necessary subject for further investigation.

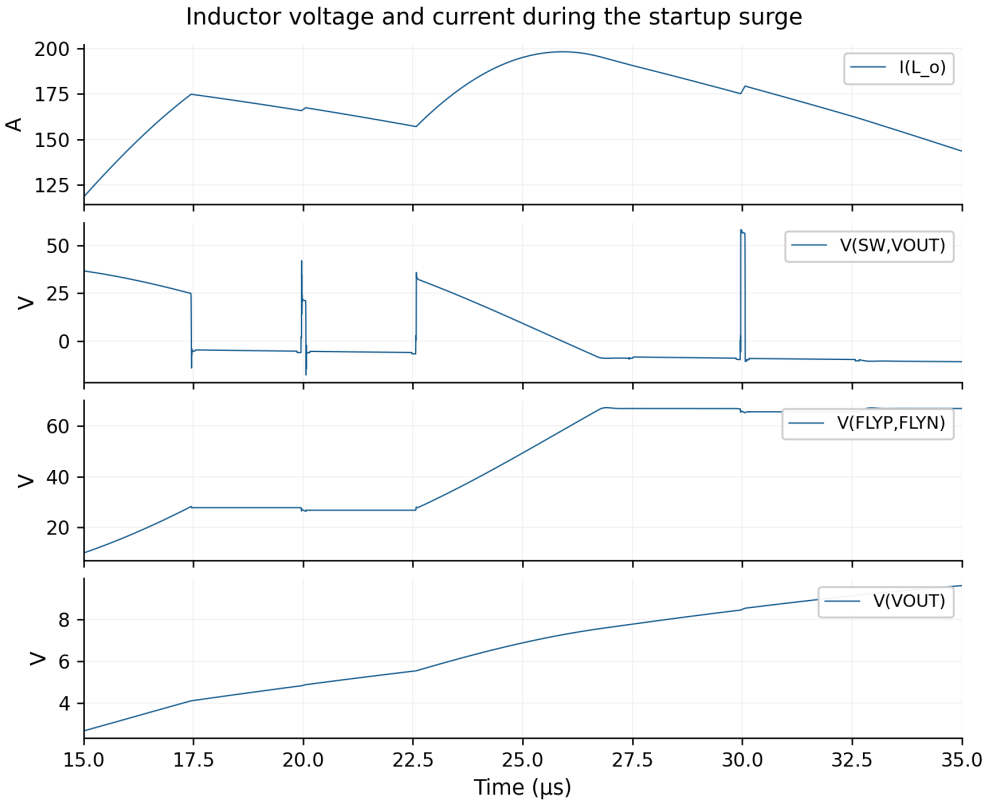


Figure 20: Inductor voltage and current with output and flying-capacitor voltages during reference-ramp startup.

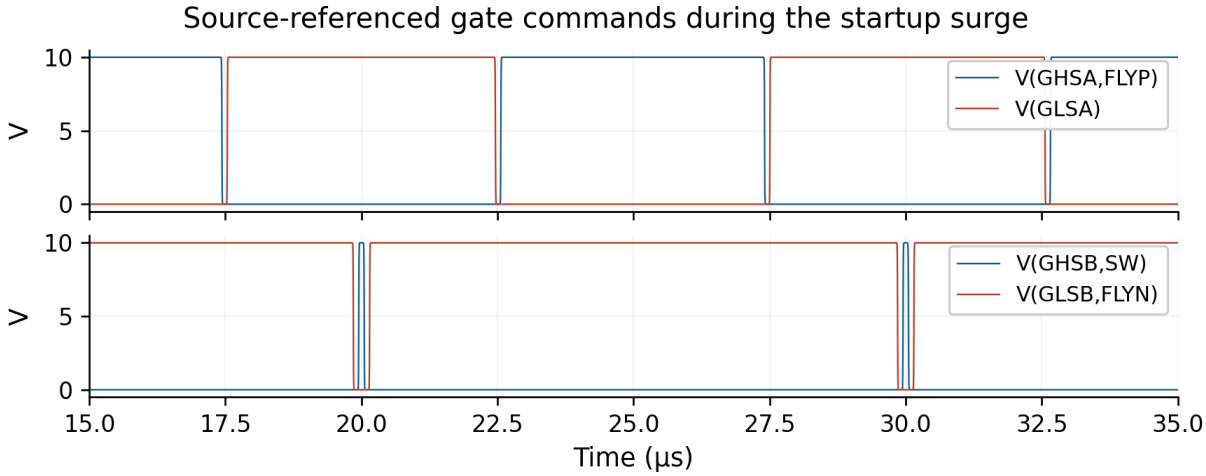


Figure 21: Four source-referenced gate commands during excessive startup current.

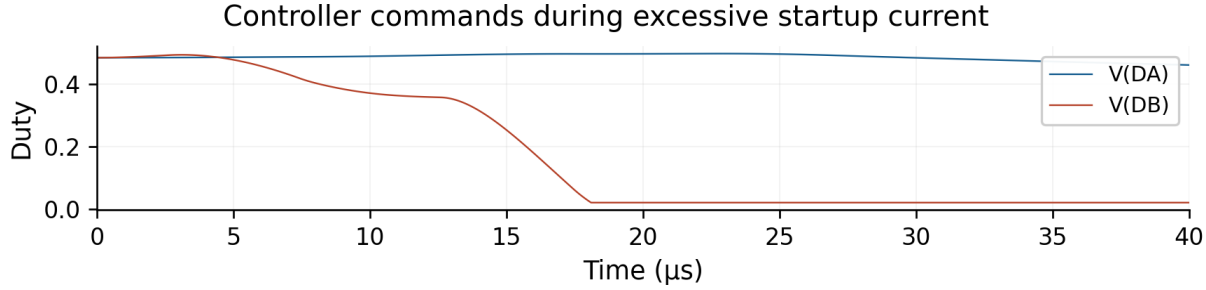


Figure 22: Flying-capacitor and output duty commands during reference-ramp startup.

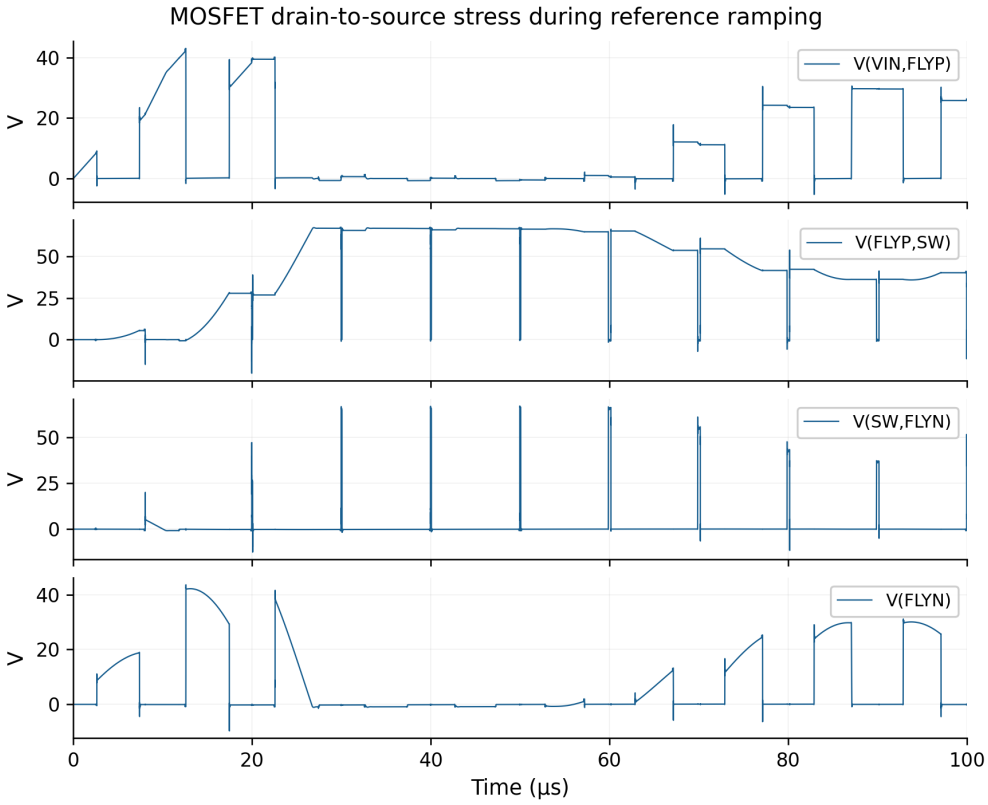


Figure 23: MOSFET drain-to-source voltages during reference-ramp startup.

6 Limitations and Scope of Reproduction

The converter and two voltage-control loops were reconstructed from the published method and model parameters. The compensator coefficients and behavioral gate drives define this LTspice implementation; they are not an exact reproduction of the authors' analog hardware. The original paper's hardware operating points and load responses provide context[1], while the results here describe simulation.

Prebiased regulation begins with stored capacitor and inductor energy. Successful recovery under those conditions does not establish zero-energy startup. The cold-start record ends at 58.273 microseconds, before its requested 400 microseconds. The reference-ramp trial covers only 100 microseconds of a 1 ms ramp. Both exhibit excessive current; the ramp trial also produces flying-capacitor overvoltage and exceeds the ratings of both 60 V switch banks. Later recovery remains unresolved.

The capacitor-current result is model-derived because direct branch current was not recorded. Commanded switching states use gate-voltage thresholds and do not resolve every channel or diode conduction path. Tolerance-relaxation warnings in the ESR comparison and solver-assisted convergence during reference ramping also limit numerical confidence. The device models have a fixed case-temperature boundary; the study establishes neither hardware safe operation nor efficiency and cooling performance.

7 Conclusion

The reconstructed three-level buck converter regulates a prebiased output near 32 V at the tested input voltages and responds to load changes. Feedback reduces late output error and ripple compared with the fixed-PWM implementation, although its initial current peak is larger.

Startup remains unsuccessful. The incomplete cold-start test reaches 580.63 A, while output-reference ramping produces a 198.21 A peak and 67.23 V flying-capacitor voltage. After the output duty reaches its lower limit, the other controller keeps Mode 1 active. The positive net inductor voltage predicts the measured current increase closely. Both 60 V switch banks exceed their ratings. Further work should evaluate coordinated startup energy limitation and flying-capacitor balancing before hardware implementation.

References

[1] J.-B. Jeong, C.-G. Kim, J.-I. Kang, and S.-K. Han, "Two Independent Single-Loop Voltage Mode Control Method for 3-Level Buck Converter," *IEEE Access*, vol. 12, pp. 151382–151394, 2024, doi: 10.1109/ACCESS.2024.3476409.

[2] Infineon Technologies, *BSC034N10LS5, OptiMOS 5 Power-Transistor, 100 V*, final datasheet, rev. 2.4, 11 October 2022. `https://www.infineon.com/assets/row/public/documents/24/49/infineon-bsc034n10ls5-datasheet-en.pdf`.

[3] Infineon Technologies, *BSC027N06LS5, OptiMOS Power-Transistor, 60 V*, final datasheet, rev. 2.3, 8 November 2023. `https://www.infineon.com/assets/row/public/documents/24/49/infineon-bsc027n06ls5-datasheet-en.pdf`.

[4] Infineon Technologies, *OptiMOS5 100V LTspice Library*, bundled file `OptiMOS5_100V_LTSpice.lib`, version 041121. Header, model notice and BSC034N10LS5 declaration inspected locally.

[5] Infineon Technologies, *OptiMOS 60V / OptiMOS5 60V LTspice Library*, bundled file `OptiMOS_OptiMOS5_60V_LTSpice.lib`, version 100122. Header, model notice and BSC027N06LS5 declaration inspected locally.

[6] Analog Devices, *LTspice Reference: Simulation Commands*, transient-analysis modifiers, official reference repository, accessed 28 October 2026. `https://github.com/analogdevicesinc/ltspice-reference/blob/main/ai_ref/SIMULATION-COMMANDS-REFERENCE.md`.

A Figure attribution

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B Model-derived flying-capacitor current

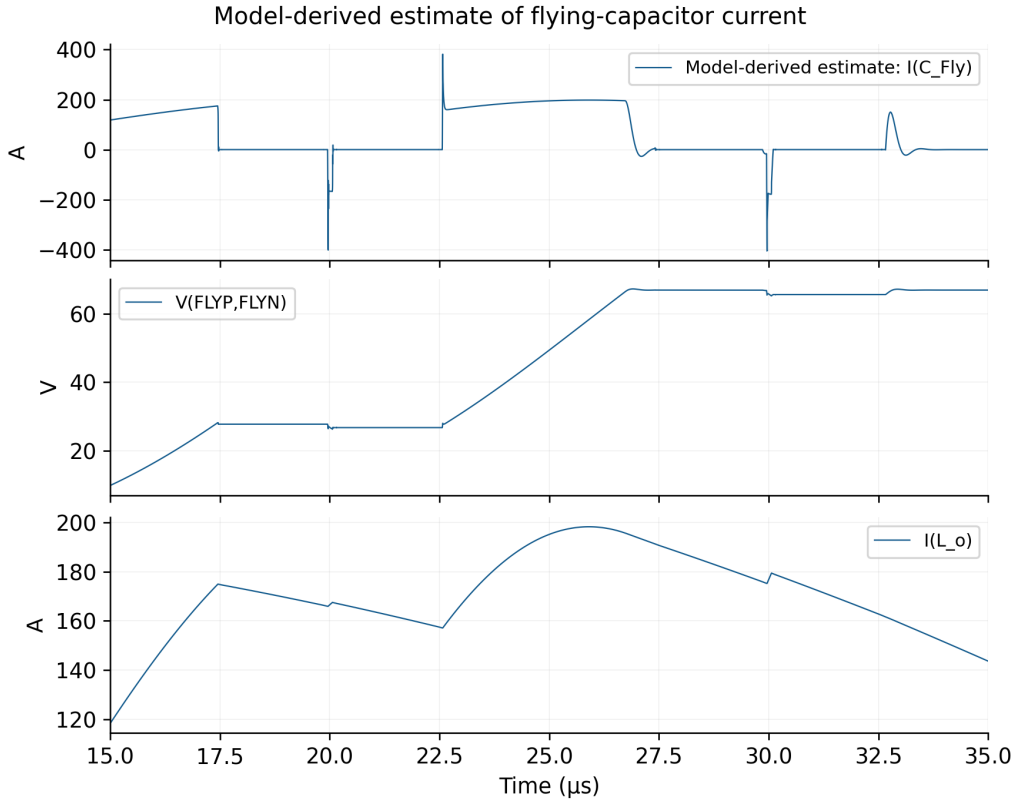


Figure 24: Model-derived flying-capacitor current alongside terminal voltage and inductor current during reference ramping.